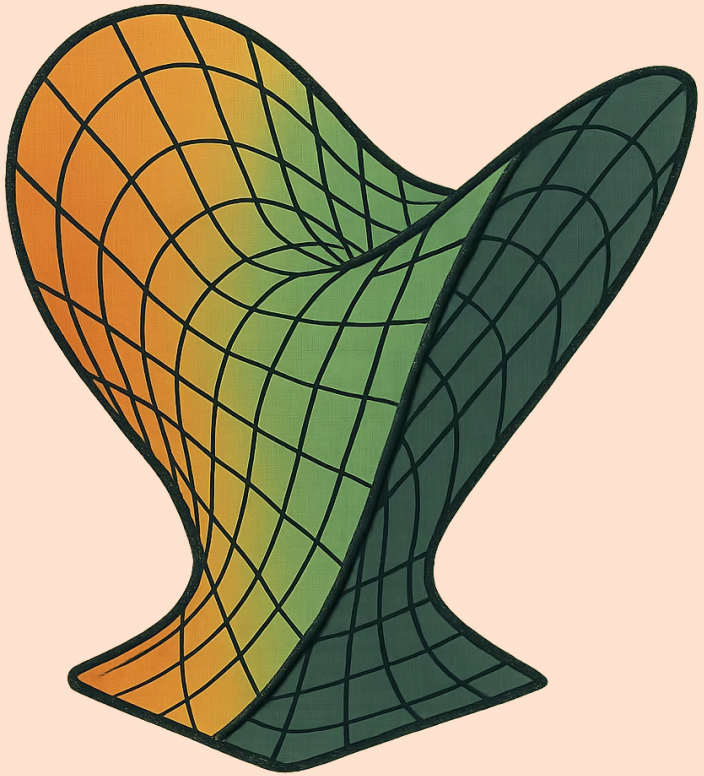

Multivariable Calculus: Integration

A rigorous exploration of multiple integrals, coordinate transformations, and their engineering applications — designed for B.Tech students.

ENGINEERING MATHEMATICS

B.TECH CURRICULUM

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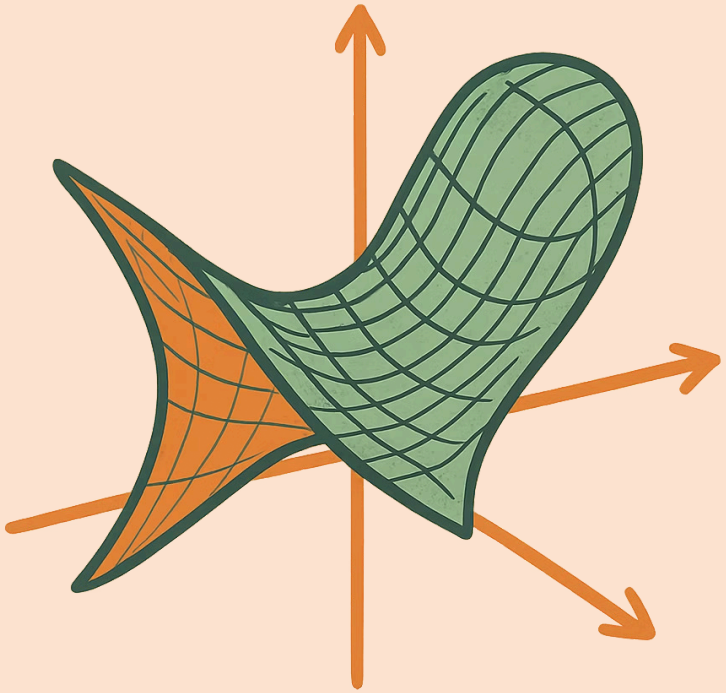

Unveiling the World of Multiple Integrals

Single-variable integration measures area under a curve. **Multiple integrals** extend this to measure volume, mass, and flux across two or three dimensions — the mathematical language of real-world engineering problems.

From 1D → 2D

Area under a curve becomes volume under a surface

From 2D → 3D

Volume becomes mass, charge, or probability in space

Real-World Relevance

Heat flow, fluid dynamics, electromagnetism, and structural analysis

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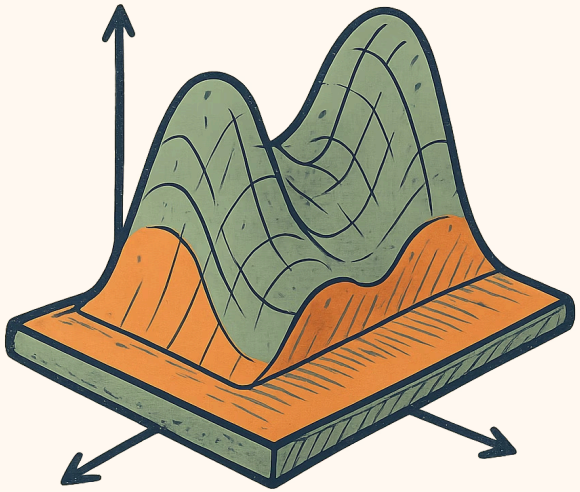
Double Integrals: Area and Volume in 2D



A **double integral** integrates a function $f(x, y)$ over a region R in the xy-plane, computing the signed volume between the surface and the plane.

Definition: $\iint_R f(x, y)\, dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i^*, y_j^*)\, \Delta A$

- **Region R** can be rectangular, circular, or any bounded domain
- If $f(x, y) = 1$, the integral yields the **area** of R
- Used to find **centroids**, moments of inertia, and surface areas

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Iterated Integrals and Fubini's Theorem

Fubini's Theorem allows us to evaluate a double integral as two successive **single integrals**, provided the function is continuous over a rectangular region.

Set Up Integral

Write double integral over region R

Apply Fubini

Express as an iterated integral

Evaluate Iteratively

Integrate inner then outer variable

This sequential approach reduces a 2D problem into manageable 1D computations.

Rectangular Region

$\int_a^b \int_c^d f(x, y)\, dy\, dx$

General Region

$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$

Limits become functions for non-rectangular domains.

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Triple Integrals: Volume and Mass in 3D

A **triple integral** extends integration over a solid region E in three-dimensional space, enabling computation of volume, mass, and centre of mass for 3D objects.

1

Volume

$V = \iiint_E 1\, dV$

2

Mass

$m = \iiint_E \rho(x, y, z)\, dV$ where ρ is density

3

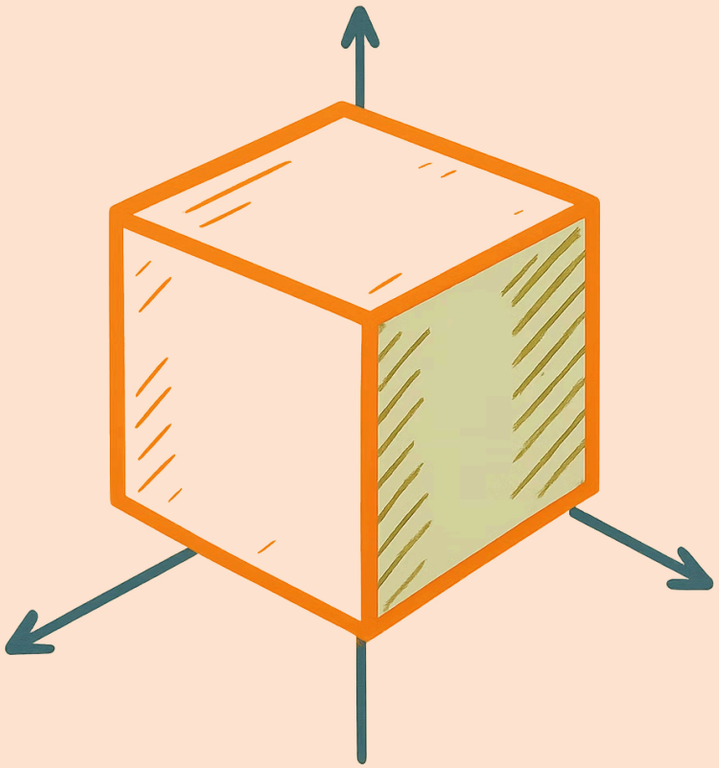
Centre of Mass

$\bar{x} = \frac{1}{m} \iiint_E x\rho\, dV$ (similarly for $\bar{y}, \bar{z}$)

4

Moment of Inertia

$I_z = \iiint_E (x^2 + y^2)\rho\, dV$



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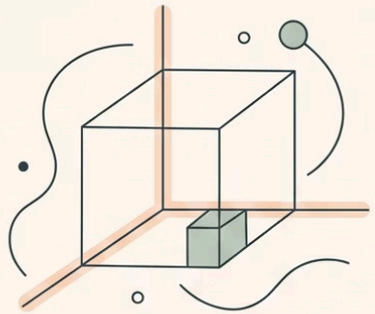
Coordinate Systems: Cartesian, Cylindrical, and Spherical

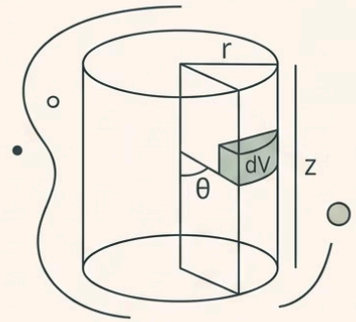


Cartesian (x, y, z)

Uses perpendicular axes.

dV = dx dy dz

Best for boxes and rectangular solids.

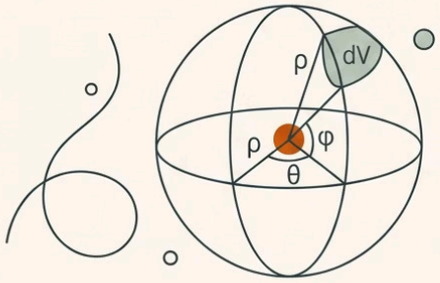


Cylindrical (r, θ, z)

Uses radius, angle, height.

dV = r dr dθ dz

Best for cylinders and pipes.



Spherical (ρ, θ, ϕ)

Uses radial distance and two angles.

$dV = \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi$

Best for spheres and cones.

Choosing the **right coordinate system** simplifies the integral dramatically. The key is matching the geometry of the region to the coordinate system.

- **Cartesian:** Boxes, rectangular plates
- **Cylindrical:** Pipes, cones, circular regions
- **Spherical:** Spheres, hemispheres, radial fields

> Always include the **Jacobian factor** (r or $\rho^2 \sin\phi$) when transforming.

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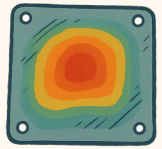
Applications of Multiple Integrals in Engineering

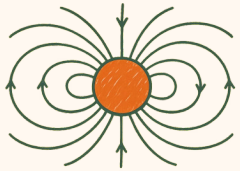


Heat & Thermodynamics

Compute total heat energy and temperature distribution across surfaces and solids using double and triple integrals.

Fluid Mechanics

Calculate mass flow rate, flux, and pressure forces in fluid dynamics and aerodynamics.



Electromagnetism

Determine electric charge, electric flux, and magnetic field properties over 3D regions.

Structural Analysis

Find centre of mass, moments of inertia, and stress distributions in mechanical components.

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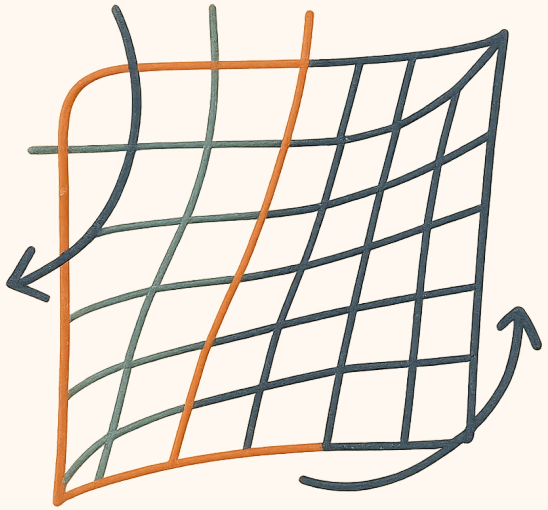
Change of Variables: Jacobians and Transformations

When transforming from (x, y) to (u, v), the **Jacobian determinant** accounts for how area or volume elements stretch or shrink under the transformation.

Jacobian Formula: $J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$

The transformed integral becomes:

$$\iint_R f(x, y)\, dx\, dy = \iint_S f(x(u, v), y(u, v))\, |J|\, du\, dv$$



Polar → Cartesian	Cylindrical → Cartesian	Spherical → Cartesian
$\lvert J\rvert = r$	$\lvert J\rvert = r$	$\lvert J\rvert = \rho^2 \sin\phi$

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Line Integrals and Surface Integrals

Line Integrals

Integrate a scalar or vector field **along a curve C** in space.

Scalar form: $\int_C f(x, y)\, ds$

Vector form (work): $\int_C \mathbf{F} \cdot d\mathbf{r}$

- Used for work done by a force field
- Path-dependent in general; path-independent for conservative fields

Surface Integrals

Integrate over a **surface S** in 3D space.

Scalar form: $\iint_S f(x, y, z)\, dS$

Vector form (flux): $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$

- Used for fluid flux through a surface
- Foundation of Gauss's and Stokes' Theorems

These concepts lead to the **Green's, Stokes', and Divergence Theorems** — the crown jewels of vector calculus.

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Key Takeaways and Further Exploration

01

Double Integrals

Compute volume, area, and centroids over 2D regions using iterated integration.

02

Triple Integrals

Extend to 3D for mass, centre of mass, and moments of inertia.

03

Coordinate Choice

Match geometry to Cartesian, Cylindrical, or Spherical coordinates for simplicity.

04

Jacobians

Use change of variables with the Jacobian determinant to simplify complex regions.

05

Line & Surface Integrals

Gateway to vector calculus theorems and advanced engineering analysis.

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